



Cloud-Scape Smart Fab: Intelligent Building Exhibition & Exchange Center

AI-Driven Computation; Parametric Form-Finding; BIM Prefabrication

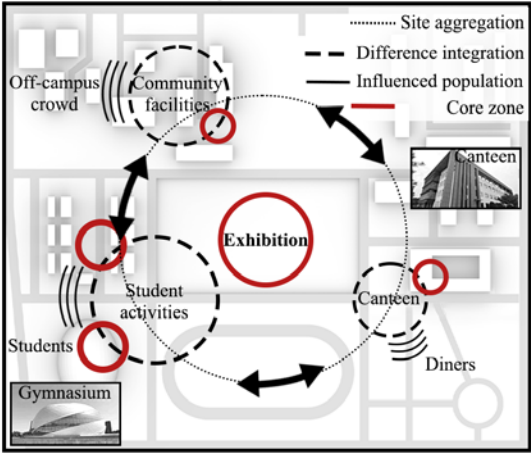
- First Prize, Beijing Building Structure Design Competition | 2025
- Interdisciplinary Project Lead

Driven by parametric algorithms, this project achieves the collaborative generation of large-span spaces and structural systems. Deeply integrating BIM platforms and finite element analysis (FEA), the design transforms complex spatial grids into standardized modular units. The scheme fully adopts prefabricated assembly technology, linking data models directly to component prefabrication and rapid on-site assembly, thereby realizing a highly efficient transition from digital morphogenesis to smart construction.

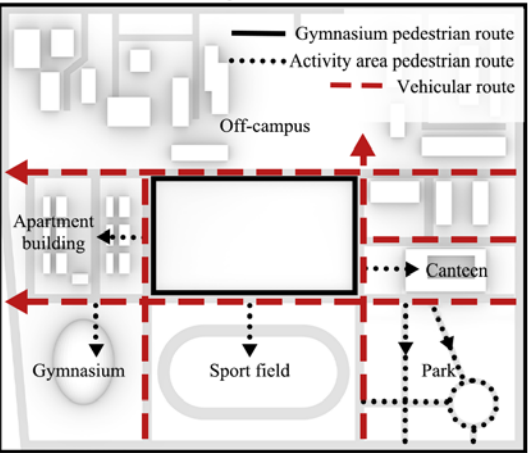
Site analysis

Taking the Intelligent Building Exhibition and Exchange Center of Beijing University of Civil Engineering and Architecture as the site selection project, this study analyzes the surrounding conditions based on site characteristics and functional requirements.

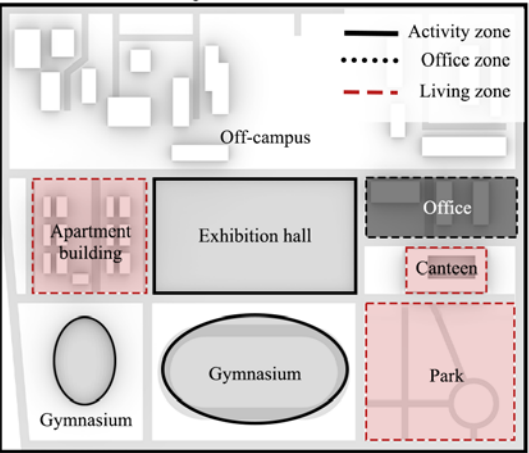
Structural analysis



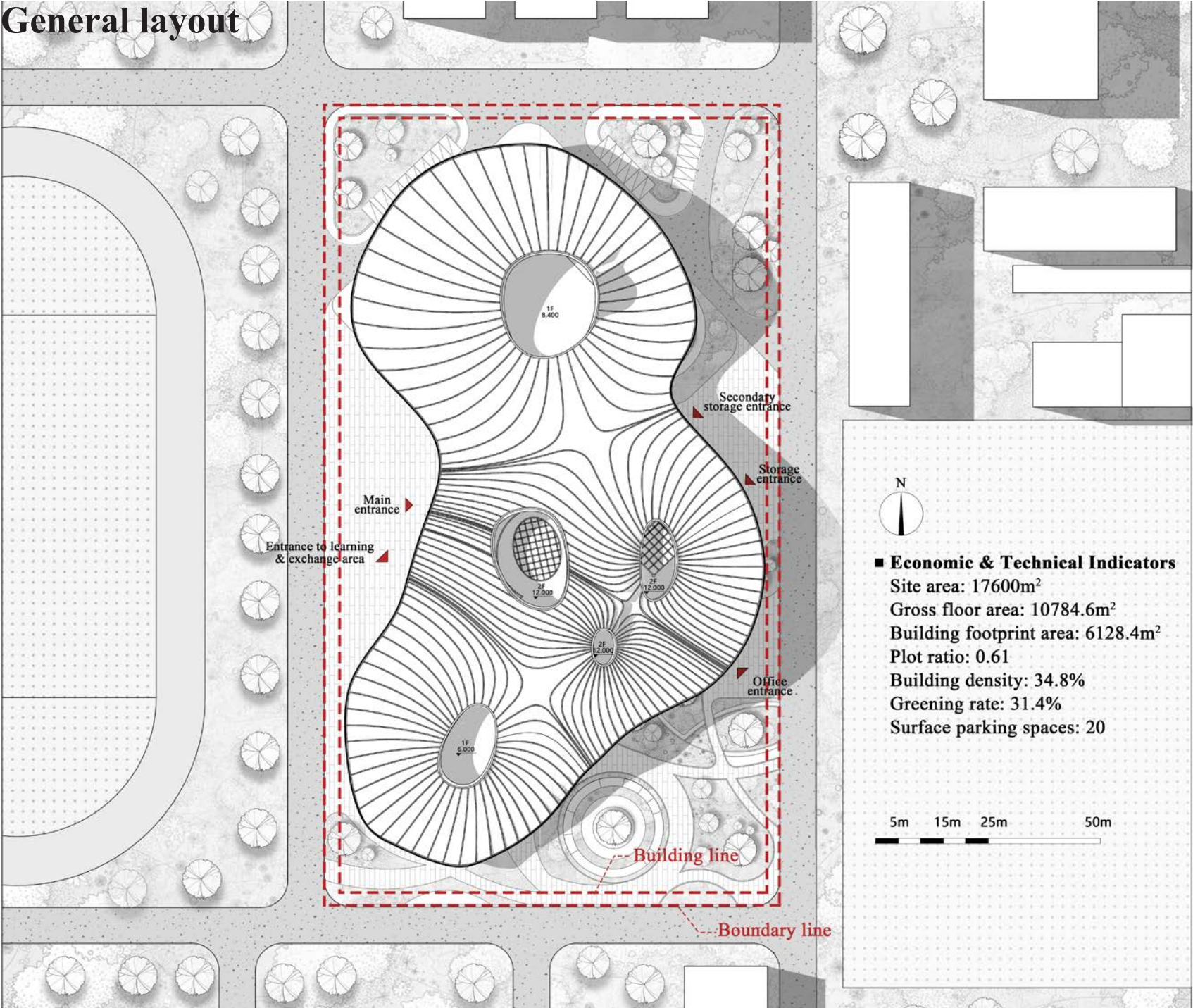
Circulation analysis



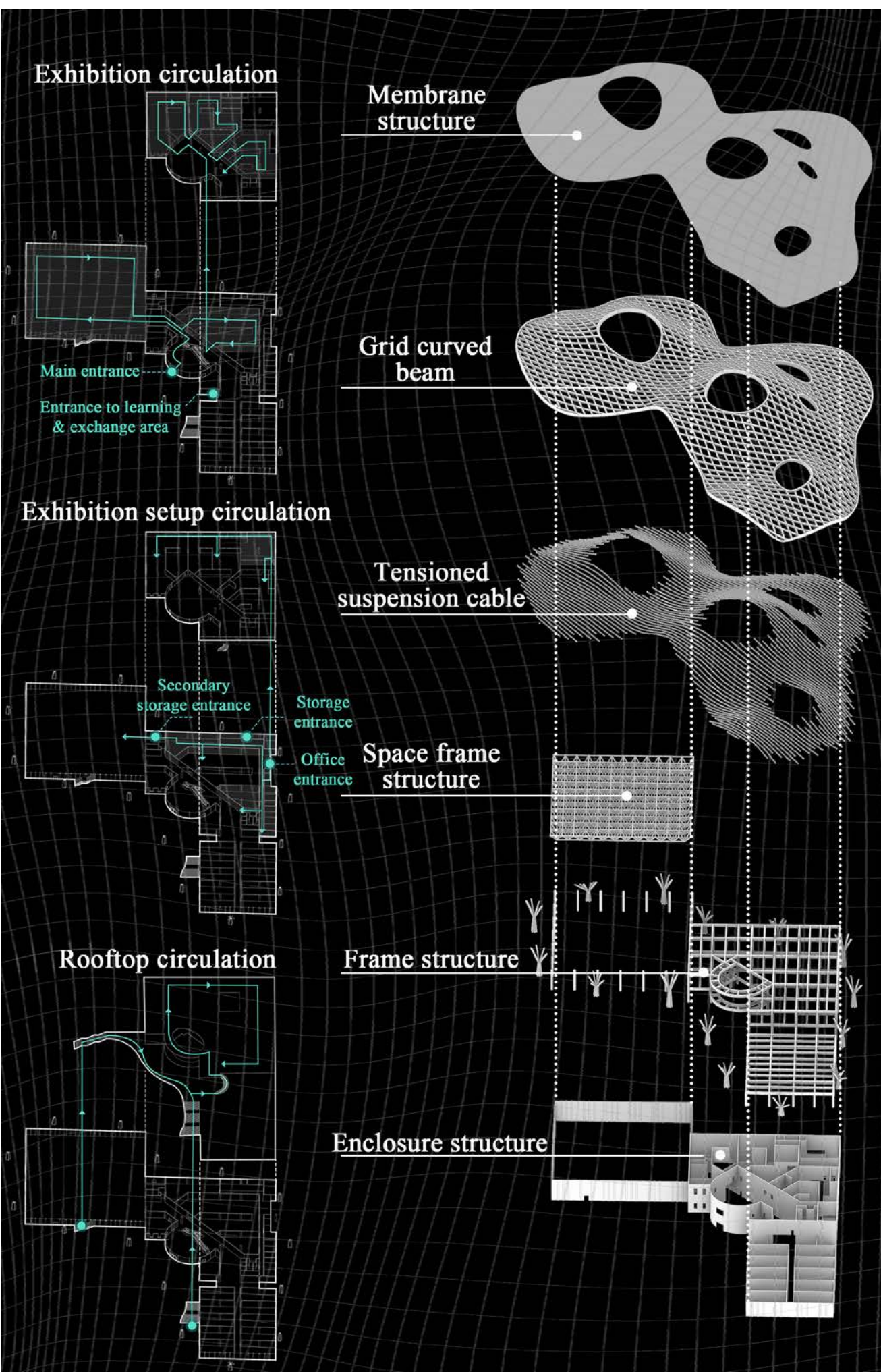
Function analysis



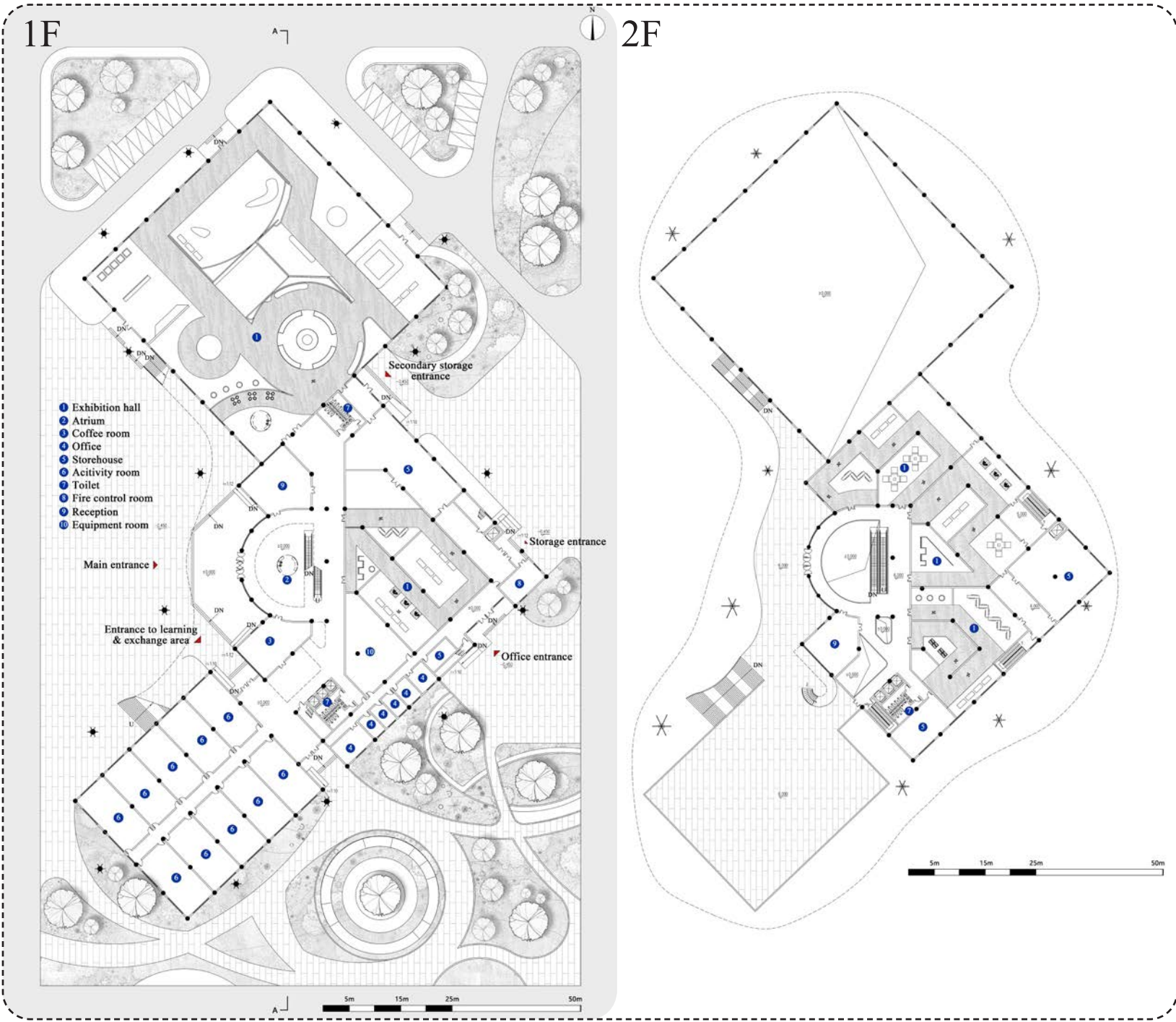
General layout



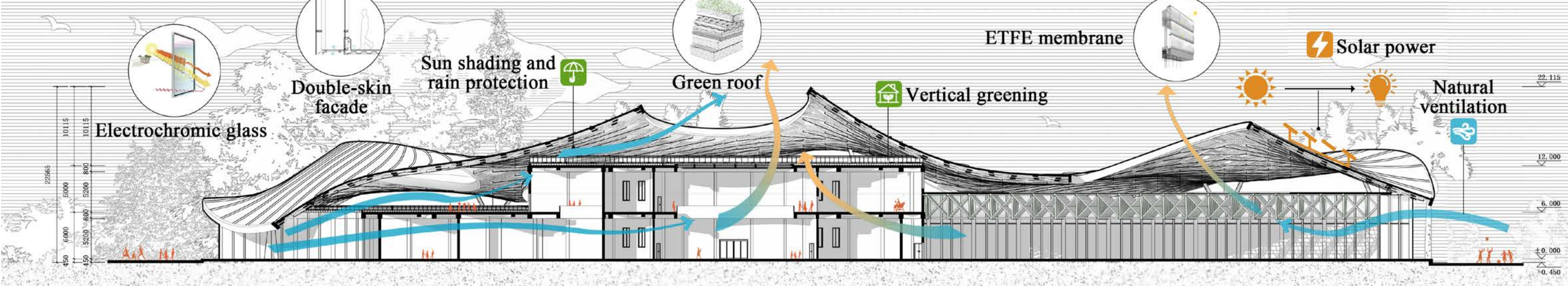
Streamline and structural analysis



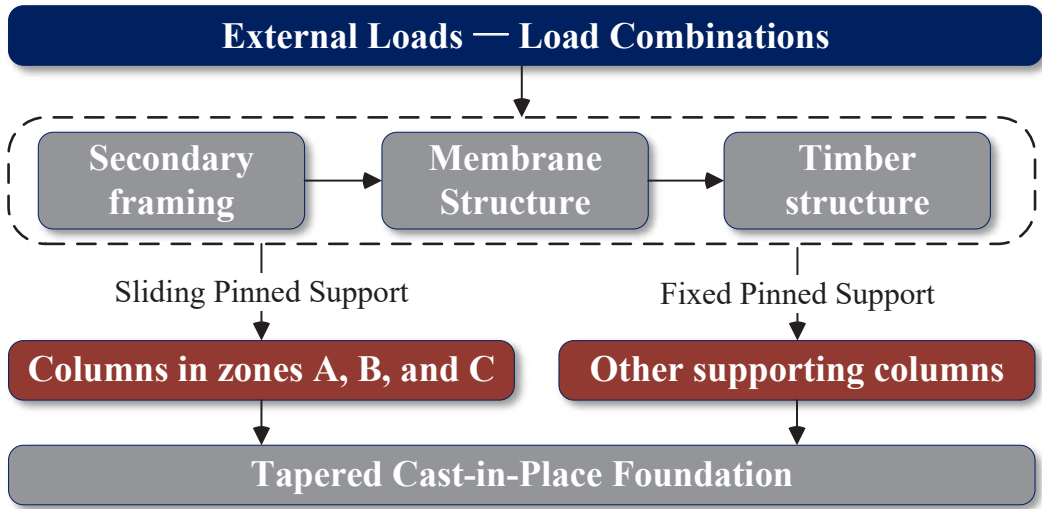
Floor plan



Section A-A



Load transfer



Structural selection

Zones B and C

These zones primarily utilize a reinforced concrete (RC) structural system, comprising frame beams, columns, and shear walls.

Zones A

This area adopts a hybrid structural system. The substructure consists of reinforced concrete supporting columns, while the superstructure features a large-span curved spatial steel structure. The roof's steel framework is composed of Q345 steel upper and lower chords, diagonal bracing, and web members (indicative of a space frame or steel pipe truss system).

Large-span roof

In addition to steel and membrane components, the large-span roof creatively incorporates a lightweight Glulam (Glued Laminated Timber) structural system.

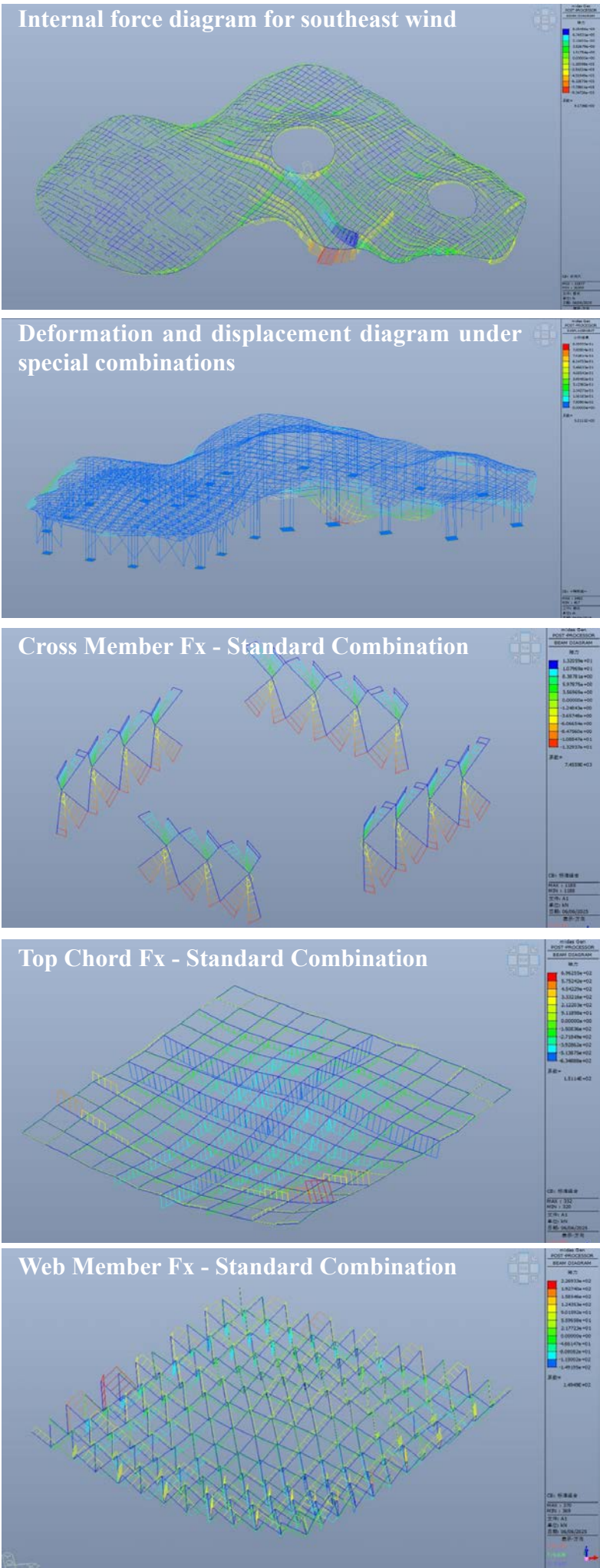
Timber beam connection

Space frame structure

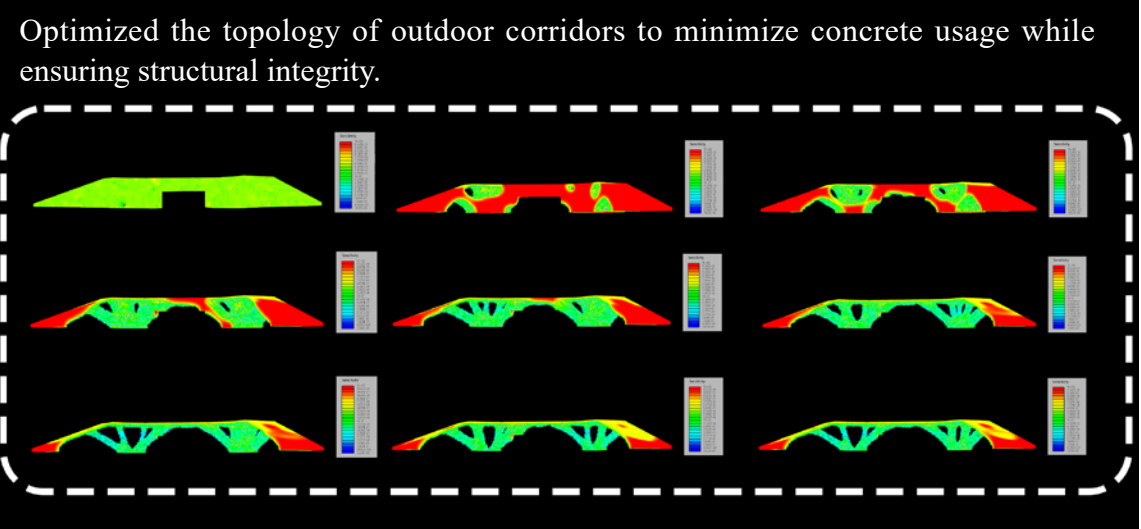
Frame structure — Beam and column

A B C

Midas structural simulation



Topology optimization of platforms



Clustering of prefabricated steel components

